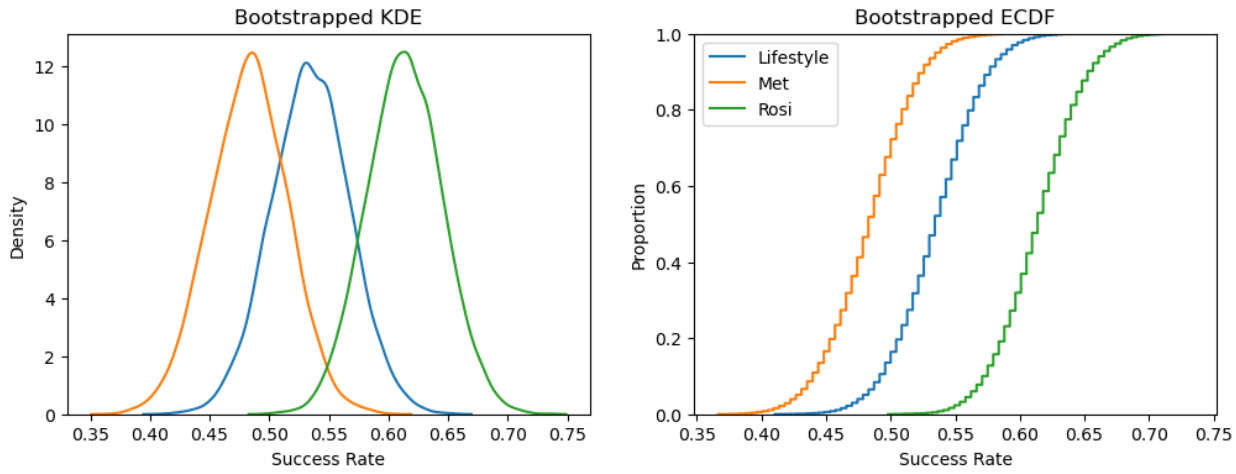
A clinical trial was conducted to consider the difference in effectiveness between treatments for pre-diabetes to reduce the progression to full diabetes. There were three key interventions considered in this trial: lifestyle adjustments, Metformin, and Rosiglitazone. Two of these, Metformin and Rosiglitazone, are prescription medications while lifestyle adjustments are making changes to diet and exercise that could reduce the progression of diabetes. For the trial, the hypothesis is that there is no difference in the success rate in reducing progression to full diabetes between the three interventions.

Looking at the average success rate of each intervention in Table 1, it appears that Rosiglitazone has the highest average success rate at 61.4% and Metformin has the lowest average success rate of 48.3%. This relationship is further shown in the bootstrap KDE and ECDF plots in Figure 1. This shows the density and distribution of the average success rate for each intervention across bootstrap samples of the data. The Bootstrapped KDE plot shows that the typical success rate among the samples for Metformin is the lowest while the typical success rate among the samples for Rosiglitazone was the highest. The Bootstrapped ECDF plot shows that this relationship is consistent throughout the data, such that for any specified proportion of the data, the success rate will be highest for Rosiglitazone and lowest for Metformin. But the KDE plot also shows that there is a large amount of overlap in the density of success rates between the three interventions.

Table 1: Average Success Rate

| **Treatment** | **Success Rate** |
|---|---|
| Lifestyle | 53.4% |
| Metformin | 48.3% |
| Rosiglitazone | 61.4% |

Figure 1: Bootstrapped Success Rate KDE and ECDF




To further investigate this relationship between the interventions, Table 3 shows the average treatment effect, or difference in success rate, between each of the three interventions. This shows that lifestyle has a 0.051 higher success rate compared to Metformin while it has a 0.08 lower success rate compared to Rosiglitazone. As well, Metformin has a 0.131 lower success rate compared to Rosiglitazone.

Table 3: Average Treatment Effect of Intervention Pairs

| **Intervention Pair** | **Average Treatment Effect** |
|---|---|
| Lifestyle vs Metformin | 0.051 |
| Lifestyle vs. Rosiglitazone | -0.080 |
| Metformin vs. Rosiglitazone | -0.131 |

While there is a difference that can be seen in the average treatment effect between each of the three interventions, this is only for one sample of the trial data. To explore how statistically sound these differences are, the bootstrap confidence intervals in Table 4 are ranges of plausible true treatment effects. If this same experiment was run on many samples, it would generate intervals that contain this true value 90%, 95%, and 99% of the time. Because zero is within the confidence interval for lifestyle versus Metformin, the null hypothesis that there is no difference between these two cannot be rejected. For lifestyle versus Rosiglitazone, it is only the case that zero is not in the 90% confidence interval. This indicates that there is a decrease in the success rate for lifestyle compared to Rosiglitazone, but it is not statistically significant at the 95% level. Lastly, every confidence interval for Metformin versus Rosiglitazone is less than zero, showing that Metformin has a statistically significantly lower success rate compared to Rosiglitazone.

Table 4: Bootstrap Confidence Intervals of Average Treatment Effect

| **Intervention Pairs** | **90% CI** | **95% CI** | **99% CI** |
|---|---|---|---|
| Lifestyle vs Metformin | -0.026 - 0.129 | -0.039 - 0.141 | -0.073 - 0.171 |
| Lifestyle vs. Rosiglitazone | -0.156 - -0.003 | -0.170 - 0.010 | -0.195 - 0.036 |
| Metformin vs. Rosiglitazone | -0.208 - -0.058 | -0.221 - -0.041 | -0.247 - -0.011 |

Overall, the results of this clinical trial testing the difference in effectiveness between treatments for pre-diabetes to reduce the progression to full diabetes are that there is no statistically significant difference between Metformin and lifestyle, lifestyle has a lower success rate than Rosiglitazone but it is only significant at the 90% level, and Metformin has a statistically significantly lower success rate compared to Rosiglitazone. This means that when choosing between prescription medication options, Rosiglitazone is most likely to be an effective choice. But, choosing between taking Rosiglitazone and making lifestyle adjustments is more complicated. While Rosiglitazone has a statistically significantly higher success rate, it is not at the 95% confidence level and has an uncertain, possibly small or large, average treatment effect magnitude. This means it will be important to consider the patient's individual circumstances, such as their ability to afford the medication, insurance, or likelihood to stick to lifestyle changes, to determine which is going to be the best choice in intervention.